

Summer Research Internship Report

Exploring GAN-based Defense Strategy for Adversarial Images using Vision Transformer

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Abstract

This report presents the work conducted during a summer research internship at the Computer Vision and Biometrics Lab, IIIT Allahabad. The project focuses on developing a robust deep learning framework against adversarial attacks using Vision Transformers (ViT) and Generative Adversarial Networks (GANs). The research explores various GAN-based defense strategies to restore the performance of Vision Transformers when subjected to adversarial perturbations. Through systematic experimentation with different loss functions and training strategies, the study demonstrates promising results in defending against Projected Gradient Descent (PGD) attacks on CIFAR-100 dataset.

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1 Introduction

Adversarial attacks pose significant challenges to deep learning models, particularly in computer vision applications. These attacks involve adding imperceptible perturbations to input images that can cause state-of-the-art models to misclassify with high confidence. Vision Transformers (ViTs), despite their superior performance on various computer vision tasks, are also vulnerable to such attacks.

This project investigates the potential of Generative Adversarial Networks (GANs) as a defense mechanism against adversarial attacks on Vision Transformers. The primary objective is to develop a framework that can restore the original image from its adversarially perturbed version, thereby maintaining the classification accuracy of the ViT model.

2 Literature Review and Background

2.1 Vision Transformers

Vision Transformers have emerged as powerful alternatives to convolutional neural networks for image classification tasks. They apply the transformer architecture, originally designed for natural language processing, to image patches treated as sequences.

2.2 Adversarial Attacks

Adversarial attacks, particularly Projected Gradient Descent (PGD), generate perturbations that maximize the loss function of the target model while keeping the perturbations imperceptible to human eyes.

2.3 GAN-based Defense Strategies

Generative Adversarial Networks can potentially serve as preprocessing modules to clean adversarial perturbations from images before they are fed to the target classifier.

3 Methodology

3.1 Dataset and Initial Setup

The experiments were conducted using the CIFAR-100 dataset, which contains 60,000 32×32 color images across 100 classes. The Vision Transformer implementation used was based on the PAEViT (Patch-Aligned Embedding Vision Transformer) architecture.

3.2 Experimental Pipeline

The research methodology was divided into multiple phases, each building upon the previous findings:

1. Baseline ViT model training
2. Adversarial attack evaluation
3. GAN-based defense development
4. Multi-objective loss function optimization
5. Alternative training strategies
6. Novel perturbation estimation approach

4 Experimental Results

4.1 Phase 1: Baseline Model Training

The Vision Transformer model was trained on CIFAR-100 dataset achieving a classification accuracy of **74.17%** on clean test images. The trained model checkpoint was saved as `best.pth` for subsequent experiments.

4.2 Phase 2: Adversarial Attack Evaluation

Projected Gradient Descent (PGD) attacks were applied to the test dataset. The evaluation revealed a dramatic drop in accuracy from 74.17% to **6.66%**, demonstrating the severe vulnerability of the ViT model to adversarial perturbations.

4.3 Phase 3: Initial GAN Training with Frozen ViT

The first defense approach involved using a frozen ViT encoder combined with a Pix2Pix decoder as the generator. Key observations include:

- Cross-Entropy loss ranged between [4.0-4.47] across batches
- Average Generator Loss: ~ 2.08
- Average Discriminator Loss: ~ 0.325
- Limited improvement due to stagnant adversarial loss

4.4 Phase 4: U-Net Based Generator

The ViT encoder in the generator was replaced with a U-Net architecture. A custom multi-objective loss function was implemented:

$$\mathcal{L}_{total} = \lambda_1 \cdot \mathcal{L}_{adv} + \lambda_2 \cdot \mathcal{L}_{mse} + \lambda_3 \cdot \mathcal{L}_{ce} \quad (1)$$

where:

- $\mathcal{L}_{adv}$: Adversarial loss using BCEWithLogitsLoss
- $\mathcal{L}_{mse}$: Mean Squared Error between generator output and target
- $\mathcal{L}_{ce}$: Cross-Entropy loss from frozen ViT predictions

4.5 Phase 5: Hyperparameter Optimization

Various combinations of λ_1 , λ_2 , and λ_3 were explored. Notably, for $\lambda_1 = 2.0$, $\lambda_2 = 1.0$, $\lambda_3 = 1.0$ with MSE replaced by MAE:

- Clean images accuracy: 5-6%
- Perturbed images accuracy: 26-27%

4.6 Phase 6: Training with Mixed Data

Both clean and perturbed training images were used as generator inputs and real samples for the discriminator:

- Clean accuracy improved to: **30-31%**
- Perturbed accuracy maintained: **26-27%**

4.7 Phase 7: Clean-Only Real Samples

Using both clean and perturbed data as generator inputs but only clean images as real discriminator samples yielded similar results:

- Clean accuracy: **30-31%**
- Perturbed accuracy: **26-27%**

4.8 Phase 8: Novel Perturbation Estimation

A new pipeline was developed where the generator learns to estimate the perturbation directly:

Given a perturbed image $I_p = I + P$, the generator estimates $P'' \approx P$, then computes $I' = I_p - P''$. The model is trained using L2 loss between P and P'' .

4.9 Phase 9: Correlation Analysis (Planned)

Future work includes analyzing correlations between low-pass and high-pass filtered versions of clean and perturbed images to identify structural differences and improve defense strategies.

5 Results Summary

Table 1: Accuracy Results Across Different Phases

Phase	Clean Accuracy (%)	Perturbed Accuracy (%)
Baseline ViT	74.17	6.66
Pix-2-Pix GAN architecture with ViT encoder	5-6	26-27
Pix-2-Pix GAN	5-6	26-27
Mixed Training Data	30-31	26-27
Clean-Only Real Samples	30-31	26-27

6 Technical Implementation

6.1 Tools and Frameworks

The following tools and frameworks were utilized:

- **Python**: Primary programming language
- **PyTorch**: Deep learning framework
- **PAEViT**: Vision Transformer implementation
- **Pix2Pix**: GAN architecture for image-to-image translation
- **NumPy, Matplotlib, Pandas**: Data manipulation and visualization

6.2 File Organization

The project maintains organized output files for different experimental configurations:

- Model checkpoints: `generator_λ1_λ2_λ3.pth`
- Accuracy logs: `test_accuracies_λ1_λ2_λ3.csv`
- Loss tracking: `loss_log_λ1_λ2_λ3.csv`
- Visualization: `plots_λ1_λ2_λ3/`

7 Discussion

7.1 Key Findings

The research demonstrates several important findings:

1. Vision Transformers are highly vulnerable to PGD attacks, with accuracy dropping from 74.17% to 6.66%
2. GAN-based defense strategies show promise in partially restoring performance
3. Training with both clean and perturbed data significantly improves clean image accuracy
4. The multi-objective loss function approach provides a flexible framework for optimization
5. Direct perturbation estimation presents a novel approach for adversarial defense

7.2 Challenges and Limitations

Several challenges were encountered during the research:

- Balancing the trade-off between clean and adversarial accuracy
- Computational complexity of training GANs with multiple objectives
- Limited improvement in some experimental phases due to training instabilities
- Need for extensive hyperparameter tuning

8 Future Work

8.1 Immediate Next Steps

1. Complete training and testing of the perturbation estimation pipeline
2. Implement and analyze correlation between frequency-filtered images
3. Optimize hyperparameters for better accuracy balance

8.2 Long-term Research Directions

1. Extend the approach to other datasets and architectures
2. Investigate ensemble-based defense mechanisms
3. Explore attention-based generators for better perturbation removal
4. Study the transferability of the defense across different attack methods

9 Conclusion

This summer research internship successfully explored the potential of GAN-based defense strategies against adversarial attacks on Vision Transformers. While the complete restoration to baseline accuracy remains challenging, the research demonstrates significant progress in developing robust defense mechanisms.

The multi-phase experimental approach provided valuable insights into different aspects of adversarial defense, from loss function design to training strategy optimization. The novel perturbation estimation approach shows particular promise and warrants further investigation.

The work contributes to the growing body of research on adversarial robustness and provides a solid foundation for future developments in this critical area of computer vision and machine learning security.